Potassium Tris(oxalato)ferrate(III): A Versatile Compound To Illustrate the Principles of Chemical Equilibria

Gabriel González*[1] and Miquel Seco

Departament de Química Inorgànica, Universitat de Barcelona, Martí i Franquès 1-11, E-08028 Barcelona, Spain; *ggonzalez@iciq.es

A number of articles have appeared in this *Journal* dealing with the synthesis and photochemical reactivity of the $[Fe(C_2O_4)_3]^{3-}$ anion *(1–6)*. The potassium salt is an easy product to synthesize in an introductory course on inorganic chemistry. Typically students are required to prepare this product with the aim of improving laboratory skills and as an introduction to the synthesis of coordination compounds.

Another much less exploited aspect of the $[Fe(C_2O_4)_3]^{3-}$ anion is that it is a very useful product for exploring chemical equilibria, offering an overall picture of how ligand coordination and displacement and precipitation equilibria are interrelated. The sufficient quantity of complex obtained in the synthesis permits a new utilization of the substance to show students a practical example of the chemistry of complexes in solution. The concepts of equilibrium, stability, and solubility, and the physical constants associated with these concepts (equilibrium constant, stability constant, and solubility product) can easily be examined through certain reactions of this anion. The anion presents an ideal balance between stability and lability such that a number of reactions can be displaced in one direction or the other, depending on the quantity and properties of the reagents involved.

In our first-year inorganic chemistry laboratory, after the initial synthesis was complete, various reactions designed to highlight complex equilibrium processes were assigned and successfully completed. Two hours is sufficient time to perform all the reactions, so the complete experimental work and discussion can be accomplished in a standard laboratory session (if filtration is used instead of centrifugation, the procedure will be more time-consuming).

Experiment

Procedure

The students first synthesize the complex $K_3[Fe(C_2O_4)_3]\cdot 3H_2O$. Then they perform a series of test-tube experiments either individually or in groups of two. Specific instructions are given in the Supplementary Material.ʷ They observe each reaction and note color changes and precipitation formation.

After the experimental work is finished students are given a series of questions. These questions can be answered in the laboratory book or, as an alternative, the instructor can use these questions to lead a discussion about the equilibrium, complex stability, and solubility concepts.

Chemicals and Laboratory Materials

The necessary chemicals and laboratory materials are listed in Table 1 of the Supplementary Material.ʷ The chemicals include those required for the synthesis of $K_3[Fe(C_2O_4)_3]\cdot 3H_2O$.

It is convenient to use small bottles (50 mL) to stock the solutions, which can be refilled from a reservoir bottle with a spigot. Pasteur pipets will be used to take the solutions from these bottles to the test tubes. When a number of drops are required, Pasteur pipets will be used to measure the drops. When a number of milliliters are required, a graduated test tube will be used to measure the volume (this graduated test tube can be made using a standard test tube and a marker).

Reactions

The following reactions are performed by the students. The reactions are grouped according to concepts examined.

(a) Testing the stability of tris(oxalato)ferrate(III) in acid media, checking for its decomposition products, and reverting back to the tris(oxalato)ferrate(III) complex:

$$\underset{\mathbf{1}}{[Fe(C_2O_4)_3]^{3-}} \xrightarrow{H^+} \underset{\mathbf{2}}{[Fe(H_2O)_6]^{3+}} + H_2C_2O_4$$

$$\underset{\mathbf{2}}{[Fe(H_2O)_6]^{3+}} \xrightarrow{SCN^-} \underset{\mathbf{3}}{[Fe(SCN)_3]}$$

$$\underset{\mathbf{3}}{[Fe(SCN)_3]} \xrightarrow{F^-} \underset{\mathbf{4}}{[FeF_4]^-}$$

$$\underset{\mathbf{4}}{[FeF_4]^-} \xrightarrow{C_2O_4^{\,2-}} \underset{\mathbf{1}}{[Fe(C_2O_4)_3]^{3-}}$$

(b) Testing the stability of tris(oxalato)ferrate(III) in the presence of other complexing anions and the relative stability of several Fe(III) complexes:

$$\underset{\mathbf{1}}{[Fe(C_2O_4)_3]^{3-}} \underset{C_2O_4^{\,2-}}{\overset{F^-}{\rightleftharpoons}} \underset{\mathbf{4}}{[FeF_4]^-}$$

$$\underset{\mathbf{1}}{[Fe(C_2O_4)_3]^{3-}} \xrightarrow{SCN^-} \underset{\substack{\mathbf{5}\\ \text{(small amount)}}}{[Fe(SCN)_3]}$$

$$\underset{\substack{\mathbf{5}\\ \text{(small amount)}}}{[Fe(SCN)_3]} \xrightarrow{F^-} \underset{\mathbf{4}}{[FeF_4]^-}$$

(c) Testing the stability of tris(oxalato)ferrate(III) in basic media, checking for its decomposition products, and reverting back to the tris(oxalato)ferrate(III) complex:

$$\underset{\mathbf{1}}{[Fe(C_2O_4)_3]^{3-}} \xrightarrow{OH^-} \underset{\mathbf{A}}{Fe(OH)_3} + \underset{\mathbf{6}}{C_2O_4^{\;2-}}$$

$$\underset{\mathbf{A}}{Fe(OH)_3} \xrightarrow{H_2C_2O_4} \underset{\mathbf{1}}{[Fe(C_2O_4)_3]^{3-}}$$

$$\underset{\mathbf{6}}{C_2O_4^{\;2-}} \xrightarrow{Ba^+} \underset{\mathbf{B}}{BaC_2O_4}$$

$$\underset{\mathbf{B}}{BaC_2O_4} \xrightarrow{Fe_2(SO_4)_3} \underset{\mathbf{1}}{[Fe(C_2O_4)_3]^{3-}} + \underset{\mathbf{C}}{BaSO_4}$$

(d) Testing the solubility of tris(oxalato)ferrate(III):

$$\underset{\mathbf{1}}{[Fe(C_2O_4)_3]^{3-}} \xrightarrow{Ba^{2+}} \underset{\mathbf{D}}{Ba_3[Fe(C_2O_4)_3]_2}$$

$$\underset{\mathbf{D}}{Ba_3[Fe(C_2O_4)_3]_2} \xrightarrow{H_2O,\ \Delta} \underset{\mathbf{7}}{\{Ba^{2+} + [Fe(C_2O_4)_3]^{3-}\}}$$

$$\underset{\mathbf{7}}{\{Ba^{2+} + [Fe(C_2O_4)_3]^{3-}\}} \xrightarrow{SO_4^{\;2-}} \underset{\mathbf{1}}{[Fe(C_2O_4)_3]^{3-}} + \underset{\mathbf{C}}{BaSO_4}$$

(e) Testing the stability of tris(oxalato)ferrate(III) in front of light and checking for its decomposition products:

$$\underset{\mathbf{1}}{[Fe(C_2O_4)_3]^{3-}} \xrightarrow{[Fe(CN)_6]^{3-}} \text{no reaction}$$

$$\underset{\mathbf{1}}{[Fe(C_2O_4)_3]^{3-}} \xrightarrow{h\nu} [Fe(H_2O)_6]^{2+} + CO_2\uparrow$$

$$\xrightarrow{[Fe(CN)_6]^{3-}} Fe_3[Fe(CN)_6]_2$$

Questions

1. Identify solutions **2**–**7** and precipitates **A**–**D**.
2. Write all chemical reactions involved in each step.
3. Why does the pale-green precipitate **D** give a green solution when dissolved? Justify the difference of color between solid **D** (pale green) and solid $K_3[Fe(C_2O_4)_3]\cdot 3H_2O$ (emerald green). *[Note: It is interesting to show the students that large crystals have a deeper color because they have less crystal surface than the same quantity of smaller crystals, and so there is less diffuse light. When any crystal is crushed in a mortar, the intensity of color decreases dramatically as a result of the creation of a huge number of microcrystals. If solid **D** is observed through a microscope, we will see the characteristic aggregate of very small microcrystals. We can compare it with the solid obtained after crushing a large crystal of potassium tris(oxalato)ferrate(III).]*
4. Of the following pairs of compounds, which is the most stable of each pair?
   $[Fe(SCN)_3]$ vs $[FeF_4]^-$
   $[FeF_4]^-$ vs $[Fe(C_2O_4)_3]^{3-}$
   $[Fe(H_2O)_6]^{3+}$ vs $[Fe(SCN)_3]$
   (Note: At this level we assume that Fe^{3+} in acid aqueous solution can be considered as $[Fe(H_2O)_6]^{3+}$ for simplicity of reasoning.)
5. Of the following pairs of compounds, which is the most soluble compound of each pair at room temperature?
   $Ba_3[Fe(C_2O_4)_3]_2$ vs $BaSO_4$
   $Ba(C_2O_4)$ vs $BaSO_4$
6. Of the following compounds, which has a solubility more dependent on the temperature?
   $Ba_3[Fe(C_2O_4)_3]_2$ vs $Ba(C_2O_4)$ vs $BaSO_4$
7. Of the following pairs of compounds, which of each pair gives more free Fe^{3+} in solution?
   $Fe(OH)_3$ vs $[Fe(C_2O_4)_3]^{3-}$
   $[Fe(SCN)_3]$ vs $[FeF_4]^-$
   (Note: At this level we assume that FeO(OH) can be considered as $Fe(OH)_3$ for simplicity of reasoning.)
8. Write the redox reaction that occurs between $[Fe(C_2O_4)_3]^{3-}$ and light.
9. Should we expect the previous redox reaction rate to be affected by the vacuum?

Hazards

All chemicals used are standard in a first-year university laboratory, so there are no special hazards involved and the usual precautions must be taken. Protective clothing (lab coat) and safety glasses must be worn as a general rule. Barium and fluoride salts are toxic if swallowed or if they come into contact with the skin. 6 M HCl causes burns and is irritating to the respiratory system. Therefore, the use of disposable safety gloves is recommended when handling these reagents; furthermore, 6 M HCl should be manipulated only under a fume hood. Waste is collected for disposal by a suitable authority and a neutralization kit with a weak base (Na_2CO_3) must be handy in case of acid spills. Suggestions for substitution of the more troublesome reagents are available in the Supplementary Material.W

Summary

It is interesting to note that the students answer correctly about 80% of the questions. This indicates the ability of these experiments to provide a good reflective tool about equilibrium processes.

[W]Supplemental Material

Lab instructions for the students, supplementary student assignments, suggestions for alternative reagents, and the CAS numbers for the starting materials are available in this issue of *JCE Online*.

Note

1. Current address: Institut Català d'Investigació Química, Av. Països Catalans, s/n, E-43007 Tarragona, Spain.

Literature Cited

1. Dallinger, R. F. *J. Chem. Educ.* **1995,** *72,* 936.
2. Young, C. G. *J. Chem. Educ.* **1985,** *62,* 445.
3. Olmsted, J. *J. Chem. Educ.* **1984,** *61,* 1098.
4. Aravamudan, G.; Gopalakrishnan, J.; Udupa, M. R. *J. Chem. Educ.* **1974,** *51,* 129.
5. Brooks, D. W. *J. Chem. Educ.* **1973,** *50,* 218.
6. Johnson, R. C. *J. Chem. Educ.* **1970,** *47,* 702.
7. Yost, D. M.; Simons, J. H. *Inorg. Synth.* **1939,** *1,* 36.
8. Kolthoff, I. M.; Sandell, E. B.; Meehan, E. J.; Bruckenstein, S. *Quantitative Chemical Analysis,* 4th ed.; Macmillan: New York, 1969.
9. Sandell, E. B. *Colorimetric Determination of Traces of Metals,* 3rd ed.; Wiley-Interscience: New York, 1959.
10. Robin, M. B. *Inorg. Chem.* **1962,** *1,* 337.